Constructions of Hamiltonian graphs with prescribed parameters

Thesis

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Outline

1. Problem statement
2. Approach 1: Greedy chordal search
3. Approach 2: 2-lift construction
4. Next steps

Aim of the thesis

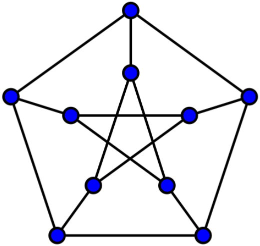
- Develop graph constructions that, for any given (k, g)-graph, produce a Hamiltonian (k, g)-graph.
- Study graphs of minimal possible order, or establish upper bounds on their order.
- Design and implement efficient algorithms for graph constructions.
- Conduct extensive computational experiments to verify graph properties.

What is a (k, g)-graph?

Definition

A (k, g)-*graph* is a simple graph that is:

- *k-regular* – every vertex has degree exactly k,
- has *girth g* – the length of its shortest cycle is exactly g.



Example of a $(3, 5)$-graph (Petersen)

Approach 1 – main idea

- **Goal:** construct a k-regular graph of girth $\geq g$ that remains Hamiltonian.
- **Initial structure:** start from a cycle C_n, which is already Hamiltonian. (where n is a Moore bound)
- **Chord addition:** add edges only between vertices that still need degree, subject to the girth constraint

$$(u, v) \text{ is allowed } \iff \operatorname{dist}_G(u, v) \geq g - 1.$$

- **Outer loop:** if k-regularity cannot be achieved, increase n (adjusted for parity if k is odd) and restart.
- **Variants:** four variants in `greedy_chordal_search/algorithms/` differ only in the rule used to select the next edge.

Approach 1 – shared greedy shell

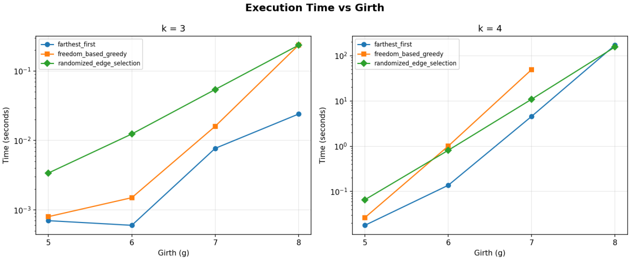
Input: k, g
Output: k-regular Hamiltonian graph with girth $\geq g$ (if found)

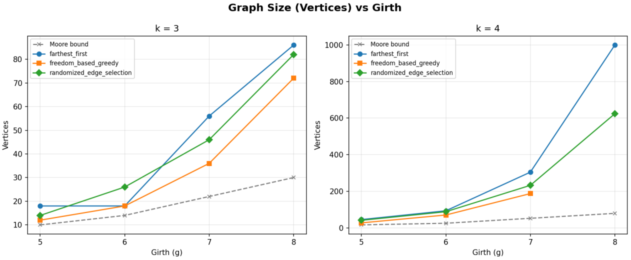
```
n ← n_0(k, g), parity-adjusted
while true do
    G ← C_n
    while there exists a girth-legal chord between unsaturated vertices do
        U ← {v : deg_G(v) < k}
        pick (u, v) using the active variant
        add edge (u, v) to G
    end
    if G is k-regular then return G

    n ← n + 1                                  // or n + 2 if k is odd
end
```

Approach 1 – edge selection variants

- **Randomized selection:** choose a random legal unsaturated pair.
- **Farthest-first heuristic:** connect vertices at maximum allowable distance.
- **Freedom-based greedy:** choose vertices with the fewest available options first.

Approach 1 – experimental results





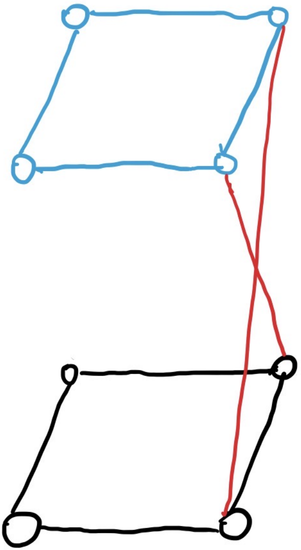
What is a 2-lift?

Given a graph $G = (V, E)$, the *2-lift* H has vertex set $V \cup V'$ (two disjoint copies of V). For each edge $uv \in E$, assign a sign $\sigma(uv) \in \{0, 1\}$:

- $\sigma(uv) = 0$ (parallel): edges uv and $u'v'$,
- $\sigma(uv) = 1$ (cross): edges uv' and $u'v$.

Properties:

- $|V(H)| = 2|V(G)|$, and regularity is preserved.
- Girth is often preserved or increased depending on σ.
- Edge choices control cycle structure.



Basic construction

Input: Connected k-regular graph G
Output: 2-lift H

Compute a 2-factor decomposition of G
Initialize all $\sigma(e) = 0$
Select one edge per cycle and set $\sigma(e) = 1$
Construct a meta-graph of cycles
Compute a spanning tree of the meta-graph
Set corresponding edges to cross edges
return resulting 2-lift H

Utilities

- For experimental testing, we used the graph database *House of Graphs*.
- Hamiltonicity checking was performed using standard computational resources, without relying on high-performance computing infrastructure (e.g., supercomputers).
- The available computational resources were sufficient for all tested graph instances.

Issue with the basic construction

Testing graphs from the *House of Graphs* database showed that the basic construction does not always produce a Hamiltonian 2-lift.

Main difficulties:

- multiple possible 2-factor decompositions,
- multiple valid spanning trees of the meta-graph,
- sensitivity of Hamiltonicity to edge selection.

More 2-factors imply a greater number of spanning trees, and consequently provide more choices for selecting cross edges.

Therefore, we tested all spanning trees for non-Hamiltonian cases.

Empirical result

Across 27 test graphs (3-regular, connected, fixed girth):

- Exhaustive spanning-tree search found a Hamiltonian 2-lift for every graph.

Next steps

- Formally characterize the optimal spanning-tree selection strategy.
- Improve constructions to achieve smaller graph orders.

Literature

L. C. Eze, R. Jajcay, T. Jajcayová, and D. Závacká, "Theoretical and computational approaches to determining sets of orders for (k, g)-graphs," *Discrete Applied Mathematics*, vol. 380, pp. 572–587, 2026.

G. Exoo and R. Jajcay, "Dynamic cage survey," *The Electronic Journal of Combinatorics*, DS16–Jul, 2012.

J. Jooken,
"Computer-assisted graph theory: a survey,"
arXiv preprint arXiv:2508.20825, 2025.

Thank you.